A Public-Data Bayesian Framework for Retrospective External Wildfire Exposure Ranking of Transmission-Line Segments

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Abstract

Transmission-line wildfire decisions often require segment-level triage, while equipment-condition, outage, and inspection records are not public. This paper presents a reproducible public-data framework for retrospective external wildfire exposure ranking of California transmission-line segments. The line segments are treated only as receptor assets. They are not modeled as ignition sources, failure points, outage causes, or responsibility units. Using California Energy Commission transmission lines, CAL FIRE FRAP fire perimeters and cause fields, gridMET weather summaries, LANDFIRE LF2024 fuel and canopy layers, and USGS 3DEP terrain, the workflow constructs a 2017–2023 segment-year panel for the Northern Sierra and Southern Cascades. The main label excludes CAL FIRE CAUSE = 11, Electrical Power, to reduce conflation between external wildfire exposure and electrical-power fire cause. A Bayesian logistic exposure model is compared with a deterministic public physics score and machine-learning baselines under a 2022–2023 temporal holdout. Bayesian Laplace and L2 logistic discrimination are close; the Bayesian contribution is therefore framed as posterior exceedance probability and posterior top-decile rank probability for rare-event triage, not as universal classification superiority. Probability calibration remains limited relative to the rare-event climatology baseline, so posterior rank probability is treated as the primary decision output.

1 Introduction

Wildfire exposure increasingly affects linear infrastructure systems. For transmission corridors, screening decisions are often made at the segment scale with incomplete public information about asset condition, protection systems, inspection history, vegetation work, and outage outcomes. A public-data workflow can therefore support transparent exposure ranking, but it should not be presented as a full operational risk model.

This study is intentionally narrow. It addresses fire-to-grid external exposure ranking for transmission-line receptor assets. It does not estimate powerline-caused ignition probability, equipment failure, outage probability, damage, or legal responsibility. Electrical-power fire overlap is retained only as a diagnostic, because a perimeter-level cause field does not identify the exact ignition device or segment.

The methodological question is also narrow. Deterministic public physics scores and machine-learning classifiers return useful point scores, but rare-event triage often asks threshold and ranking questions: which segments plausibly belong in the high-exposure tail, and how uncertain is that ranking? A Bayesian decision layer answers these questions with posterior exceedance probability and posterior top-decile rank probability. This is the main contribution, rather than a claim that Bayesian point prediction dominates all alternatives.

The contributions are:

1. a reproducible public-data segment-year exposure panel for California transmission-line receptor assets;
2. cause-screened external exposure labels that exclude electrical-power fire causes from the main validation target;
3. weather, terrain, and fuel-structure covariates for retrospective annual exposure ranking;
4. Bayesian posterior exceedance and posterior top-decile rank probabilities as decision-facing outputs; and
5. temporal-holdout comparison with deterministic, L2 logistic, tree-based machine-learning, and initial Stan MCMC baselines.

2 Study Area and Public Data

2.1 Study Area

The case study focuses on the Northern Sierra and Southern Cascades in California. Decision units are approximately 1 km transmission-line segments. The analysis panel spans 2017–2023 and uses segment-year rows so annual fire labels and annual observed environmental summaries are aligned. The current prototype uses a fixed 1 km segment buffer for perimeter-overlap labels; buffer sensitivity is a required follow-up analysis. Figure 1 shows the study region and receptor-asset segments.

2.2 Public Data Sources

Table 1 summarizes the public data sources. The processed panel contains 11,255 line segments and 78,785 segment-year observations. The environmental covariate maps are treated as covariate diagnostics and are moved to the Appendix so that the main paper emphasizes label construction, validation, and decision quantities.

Table 1: Public data sources and analytical roles.

Dataset	Source	Resolution / unit	Version or years	Role and caveat
Transmission lines	California Energy Commission	Public line geometry	Current public release	Receptor assets only; not equipment-failure data.
Fire perimeters	CAL FIRE FRAP	Perimeter polygons with cause fields	2017–2023 subset	Exposure labels; not damage or outage observations.
gridMET	University of Idaho gridMET	Daily weather, annual summaries	2017–2023	Observed annual weather covariates.
LANDFIRE	LANDFIRE LF2024	Fuel and canopy rasters	LF2024	Static fuel-structure covariates; time-matched sensitivity is required.
USGS 3DEP	USGS 3DEP	Elevation raster	Current public DEM	Terrain covariate; not a fire-spread model.

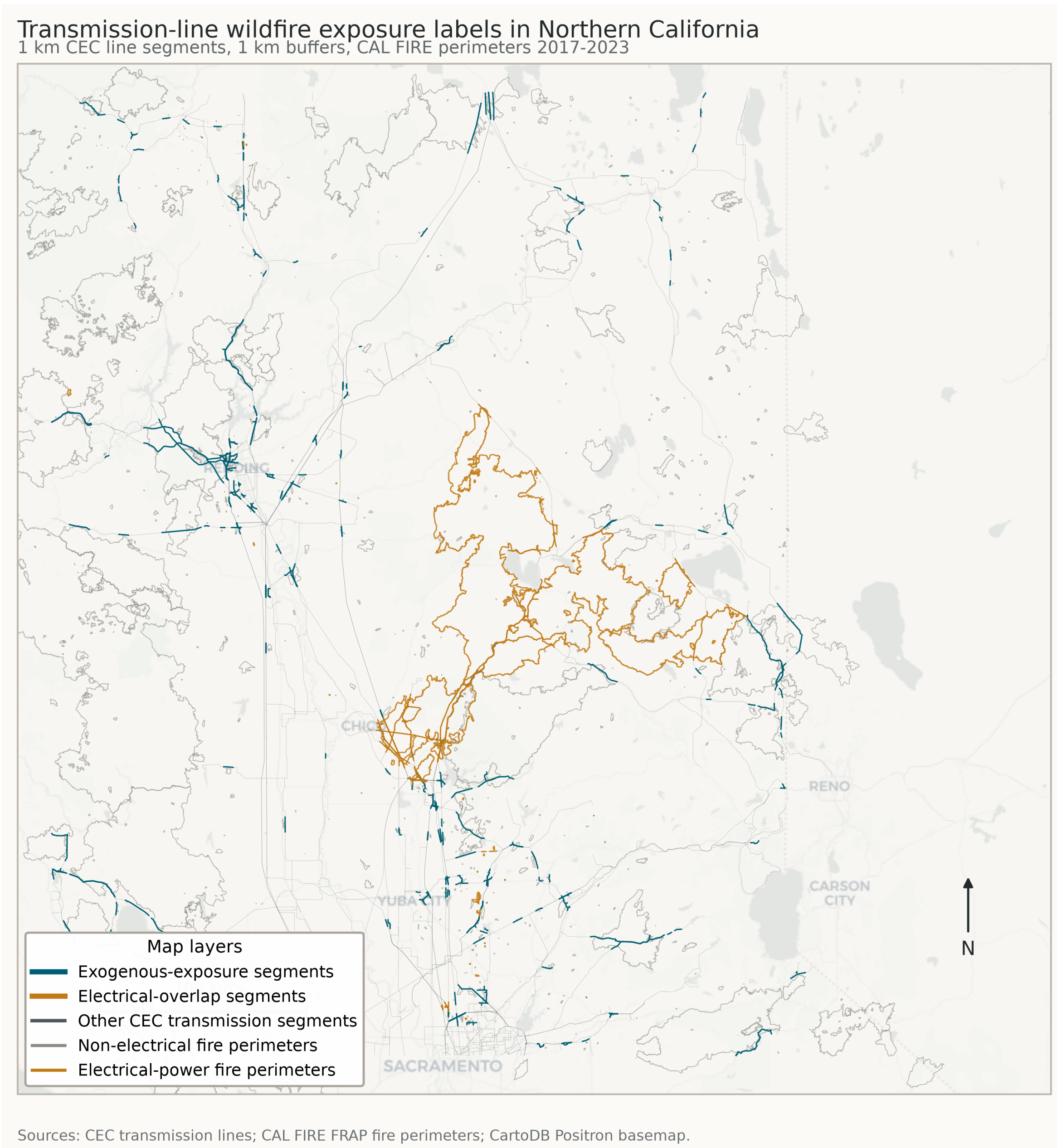

Figure 1: Study region, public transmission-line segments, and annual exposure labels.

3 Label Construction

Let i index line segments, t years, and j individual fire perimeters. Let B_i be the segment buffer and P_j be fire perimeter j. The main external exposure label is

$$y_{it}^{exo} = \mathbf{1}\left\{\exists j: \ \text{year}(j) = t, \ B_i \cap P_j \neq \emptyset, \ \text{CAUSE}_j \neq 11\right\}. \tag{1}$$

CAL FIRE CAUSE = 11 denotes Electrical Power. The exclusion in Equation 1 is used to avoid mixing external exposure with electrical-power fire cause in the main validation target. It does not assert that remaining fires cannot affect or be affected by transmission assets.

Four annual labels are retained (Table 2). The all-fire label records any perimeter overlap. The exogenous label is the main target. The strict label further excludes Equipment Use. Electrical-power overlap is a diagnostic only and is not interpreted as ignition probability.

Table 2: Annual segment-year labels. Labels are not mutually exclusive because multiple perimeters can overlap a segment-year.

Label	Definition	Positives	Interpretation
All-fire exposure	Any 1 km segment-buffer intersection with a CAL FIRE perimeter	3,187	Broad exposure screen.
Exogenous exposure	Any overlap after excluding CAUSE = 11	2,331	Main external exposure target.
Strict exogenous exposure	Excludes CAUSE in {Electrical Power, Equipment Use}	2,075	Sensitivity label.
Electrical-power overlap	Any overlap with CAUSE = 11	885	Diagnostic only.
Mixed exogenous/electrical rows	Segment-years with both exogenous and electrical-power overlaps	29	Multiple-fire overlap diagnostic.

Figure 2: Cause-screened exposure labels used for receptor-asset validation.

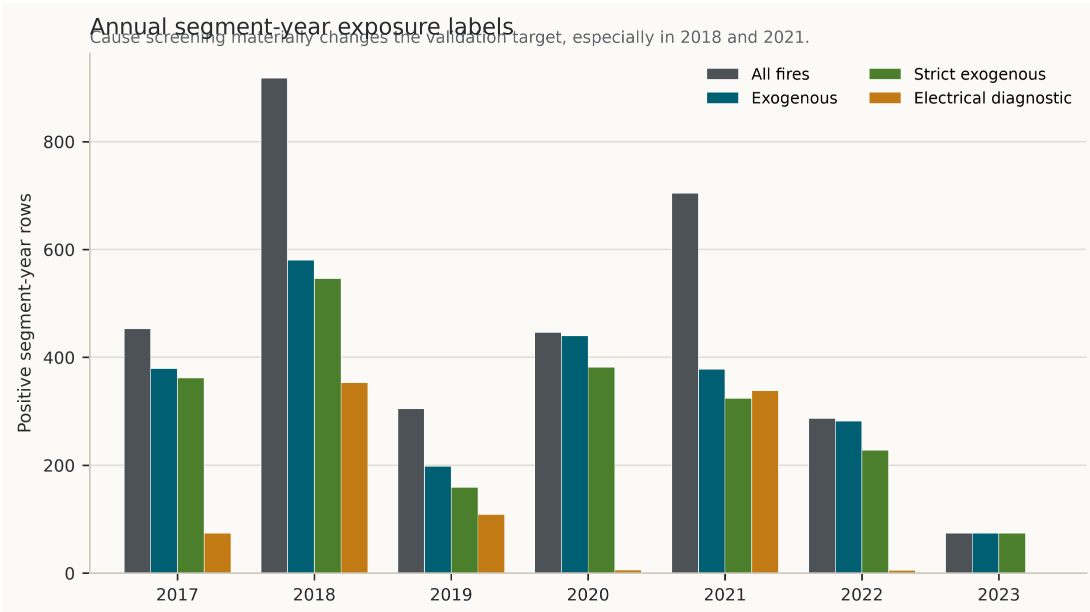

Figure 3: Annual segment-year label counts.

4 Methods

4.1 Feature Core

The selected feature core combines observed annual gridMET dryness and wind summaries, USGS terrain, LANDFIRE canopy/fuel-structure variables, and simple physics-guided indices. In this paper, physics-guided means that covariates are engineered from known weather, fuel, and terrain controls; it does not mean that the model simulates fire spread.

Let $z(\cdot)$ denote training-period standardization. The dryness index is

$$d_{it} = \frac{z(\mathrm{VPD}_{it}) + z(\mathrm{ERC}_{it}) + z(\mathrm{BI}_{it}) + z(\mathrm{wind}_{it})}{4} - \frac{z(\mathrm{FM100}_{it}) + z(\mathrm{FM1000}_{it}) + z(\mathrm{RMIN}_{it}) + z(\mathrm{PR}_{it})}{4}. \tag{2}$$

The fuel-structure index is

$$f_i = \frac{z(\mathrm{canopy\ cover}_i) + z(\mathrm{canopy\ height}_i) + z(\mathrm{canopy\ bulk\ density}_i)}{3}. \tag{3}$$

The selected core adds $d_{it}f_i$ to the raw public covariates. LANDFIRE FBFM40 is categorical and is not used in the selected Bayesian or L2 logistic core. The current Phase 5 ablation scripts used FBFM40 as a continuous diagnostic variable in several ablations; those ablations are therefore reported as exploratory until categorical encoding is implemented.

4.2 Deterministic Comparator

The deterministic comparator is a transparent, calibrated physics-guided score. It is a benchmark, not the final decision model:

$$r_{it}^{det} = 0.56d_{it} + 0.30f_i + 0.14z(\mathrm{elevation}_i). \tag{4}$$

Table 3: Feature-construction audit for the current revision.

Component	Current implementation	Revision status
gridMET annual weather	Annual summaries for VPD, ERC, BI, wind, fuel moisture, humidity, and precipitation	Interpreted retrospectively under observed annual conditions.
USGS 3DEP terrain	Segment midpoint elevation and derived terrain covariates	Suitable for screening; not a spread simulation.
LANDFIRE canopy layers	CH/CBH/CBD sampled as raw raster-coded values in processed tables	Report decoded units: CH/CBH raw/10, CBD raw/100.
LANDFIRE FBFM40	Used as continuous in some exploratory ablations, not in selected core	Must be categorical or removed in journal-grade ablations.
Engineered physics indices	Dryness, fuel structure, dryness-by-fuel interaction	Constant unit rescaling does not change standardized linear fit but does change interpretation.

The raw score is clipped to the training-period 1st–99th percentile range and scaled to $[0, 1]$. It is then calibrated with a regularized logistic model,

$$p_{it}^{det} = \operatorname{logit}^{-1}(\gamma_0 + \gamma_1 s_{it}^{det}). \tag{5}$$

This gives a single score and calibrated point probability but no posterior distribution over ranks.

4.3 Bayesian and L2 Logistic Models

The Bayesian exposure model is a weakly regularized logistic regression:

$$y_{it}^{exo} \sim \text{Bernoulli}(p_{it}), \tag{6}$$

$$p_{it} = \operatorname{logit}^{-1}(\alpha + \mathbf{x}_{it}^{\top}\boldsymbol{\beta}), \tag{7}$$

$$\alpha \sim \mathcal{N}(0, 5^2), \qquad \beta_k \sim \mathcal{N}(0, 2^2). \tag{8}$$

The posterior is approximated by a Laplace approximation around the posterior mode $\hat{\theta}$, where

$$p(\theta \mid D) \approx \mathcal{N}\left(\hat{\theta},\ H(\hat{\theta})^{-1}\right), \tag{9}$$

and $H(\hat{\theta})$ is the negative Hessian of the log posterior. This is closely related to L2 logistic regression: a Gaussian prior on coefficients is equivalent to an L2 penalty at the posterior mode. Thus similar discrimination between Bayesian Laplace and L2 logistic is expected. The Bayesian layer is retained because posterior draws support rank and threshold decision quantities.

For posterior draw s, segment exposure over holdout years $T^\star$ is

$$\bar{p}_i^{(s)} = \frac{1}{|T^\star|} \sum_{t \in T^\star} \operatorname{logit}^{-1}\left(\alpha^{(s)} + \mathbf{x}_{it}^{\top}\boldsymbol{\beta}^{(s)}\right). \tag{10}$$

The decision quantities are

$$q_i = P(\bar{p}_i > \tau \mid D) \approx \frac{1}{S}\sum_{s=1}^{S} \mathbf{1}\{\bar{p}_i^{(s)} > \tau\}, \tag{11}$$

$$\rho_i = P(\mathrm{rank}(\bar{p}_i) \le K \mid D) \approx \frac{1}{S}\sum_{s=1}^{S} \mathbf{1}\{\mathrm{rank}(\bar{p}_i^{(s)}) \le K\}. \tag{12}$$

Here τ is a prevalence-derived exposure threshold and K is the top-decile segment count. Because Brier skill is not positive relative to the holdout climatology baseline, ρ_i is emphasized as the primary decision quantity and q_i as a secondary threshold diagnostic.

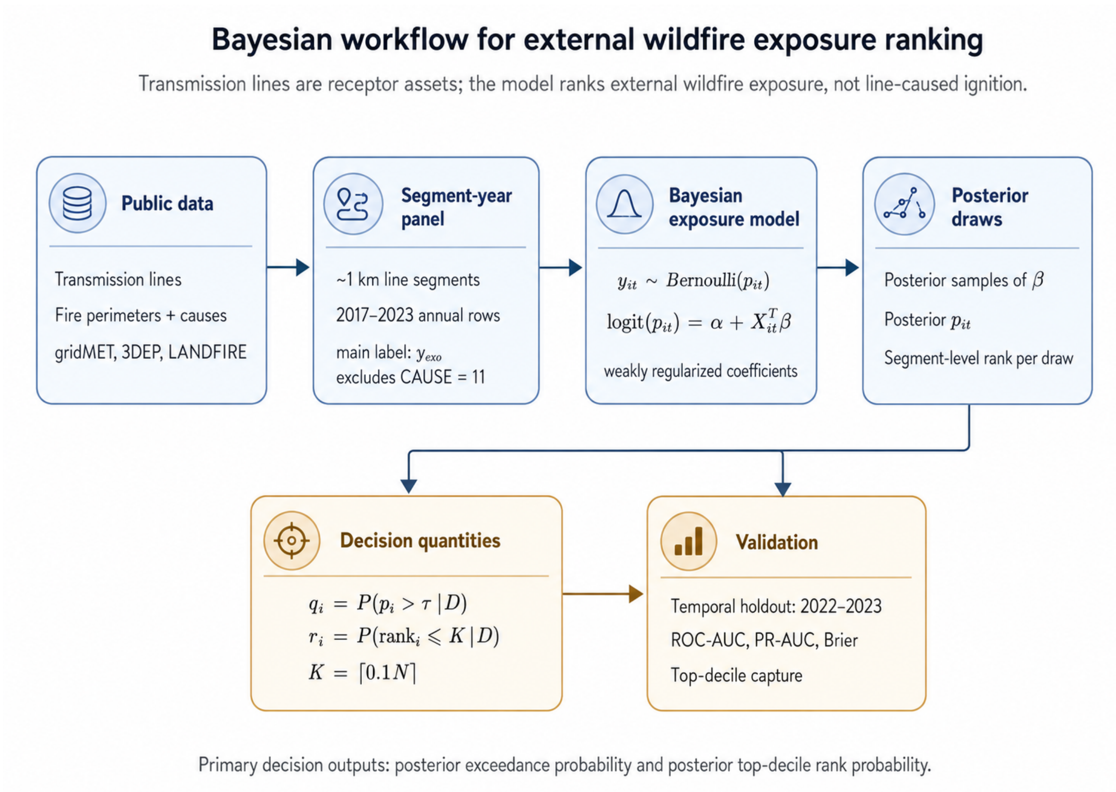

Figure 4: Bayesian workflow for posterior exposure and rank probabilities.

4.4 Validation Design

Models are trained on 2017–2021 and evaluated on 2022–2023. The holdout set contains 22,510 segment-year rows and 356 exogenous exposure positives, giving a prevalence of 1.5815%. PR-AUC is therefore interpreted relative to this rare-event prevalence. Metrics include ROC-AUC, PR-AUC, Brier score, Brier skill versus climatology, row-level top-k capture, segment-level top-decile capture, and a preliminary segment-cluster bootstrap. Temporal holdout is not a substitute for spatial block validation.

5 Results

5.1 Temporal Holdout Performance

Table 4 summarizes temporal-holdout performance. Bayesian Laplace, Stan MCMC pilot, and L2 logistic have similar discrimination. Relative to the deterministic comparator, Bayesian/L2 ranking captures more holdout positives in the top-ranked tail. Relative to L2 logistic, the Bayesian point prediction is not materially better; its value is the posterior decision layer.

Table 4: Temporal-holdout comparison for 2022–2023 exogenous exposure.

Model	ROC-AUC	PR-AUC	Brier	Row top 10%	Segment top 10%
ML logistic L2	0.663	0.039	0.015713	0.289	0.216
Bayesian Laplace	0.662	0.038	0.015718	0.289	0.216
Stan MCMC pilot	0.662	0.038	0.015710	0.289	0.216
Deterministic calibrated	0.651	0.028	0.015734	0.219	0.118
Balanced extra trees	0.617	0.022	0.056091	0.180	0.104
Histogram gradient boosting	0.553	0.019	0.016647	0.132	0.149
Balanced random forest	0.568	0.018	0.034539	0.107	0.152

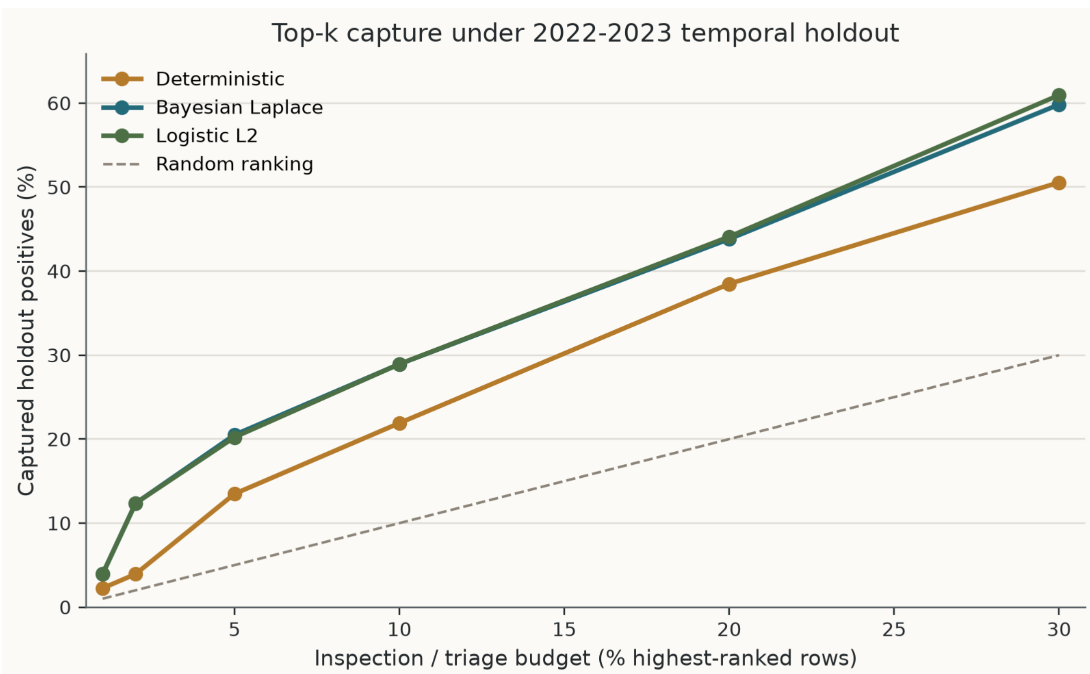

Figure 5: Top-k capture curve under 2022–2023 temporal holdout. Bayesian and L2 logistic rankings are close and both exceed the deterministic comparator in this split.

5.2 Posterior Decision Maps

The posterior map outputs should be read as uncertainty-aware exposure-ranking diagnostics. In particular, $\rho_i = P(\text{rank} \leq K \mid D)$ identifies segments that repeatedly fall into the top decile

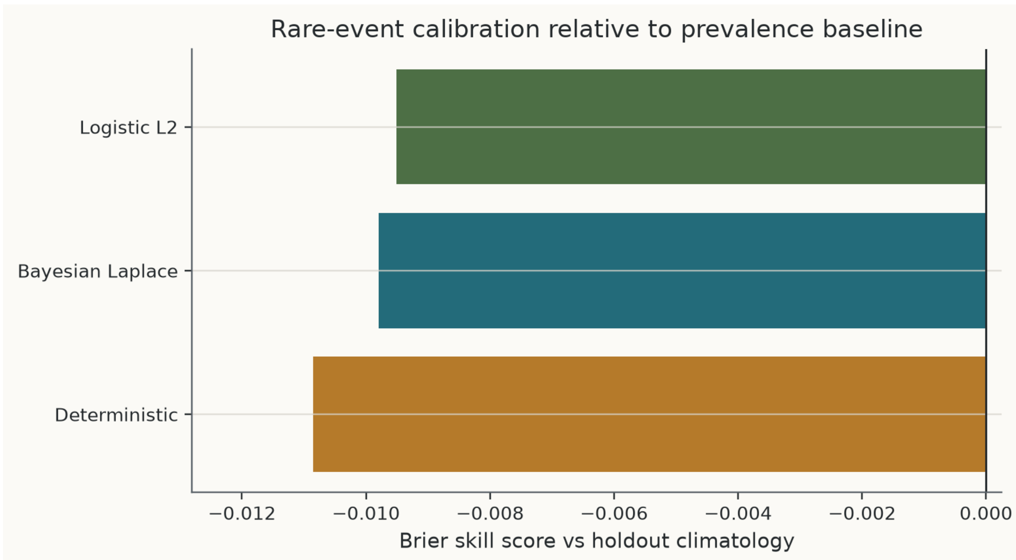

Figure 6: Brier skill relative to the holdout climatology baseline. Negative values motivate cautious interpretation of calibrated probabilities.

Table 5: Rare-event baseline audit. Holdout prevalence is 1.5815%, and the climatology Brier score is 0.015565.

Model	PR-AUC	PR-AUC / prevalence	Brier	Brier skill
Logistic L2	0.03855	2.438	0.015713	-0.0095
Bayesian Laplace	0.03838	2.427	0.015718	-0.0098
Deterministic calibrated	0.02790	1.764	0.015734	-0.0109

under posterior draws. Figure 7 and Figure 8 show the posterior decision metrics for the current prototype.

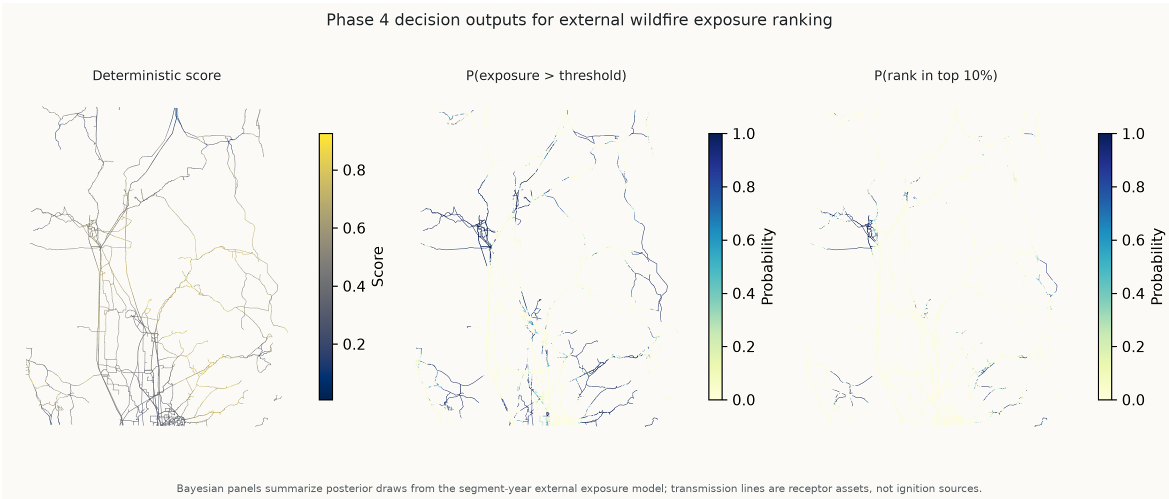

Figure 7: Posterior decision metrics mapped to transmission-line segments.

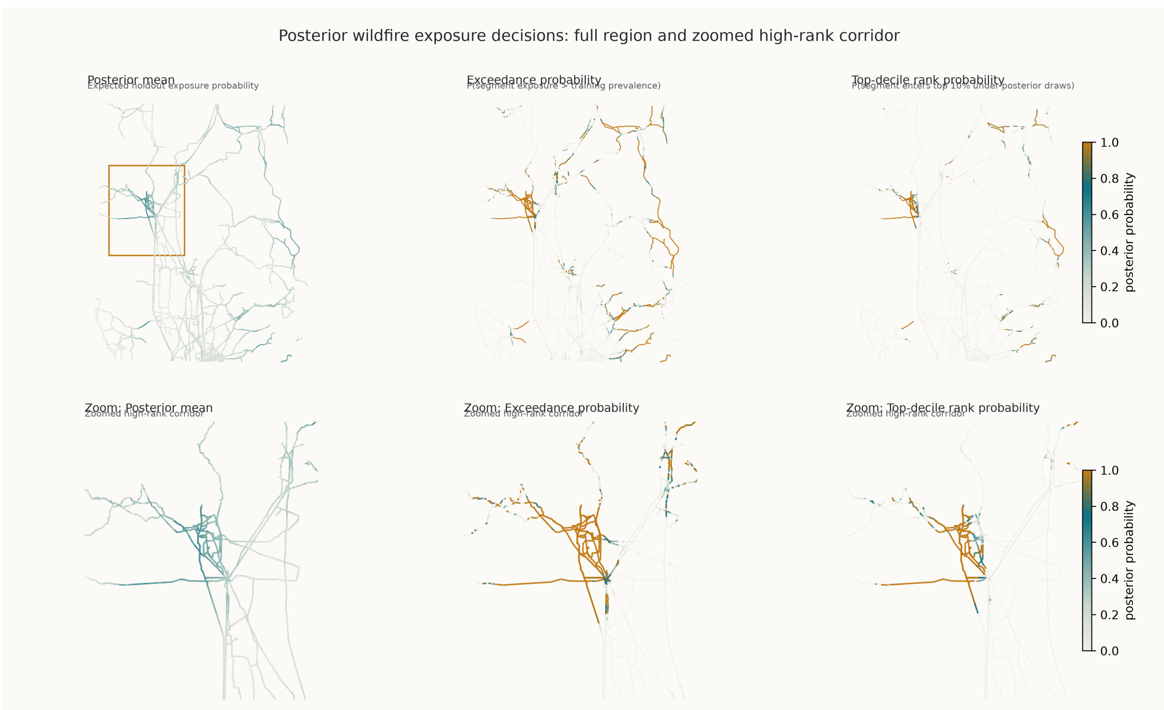

Figure 8: Full-region and zoomed maps of posterior rank and exceedance diagnostics.

5.3 Feature and Method Audits

Figure 9 shows the LANDFIRE unit audit. The selected Bayesian and L2 logistic core does not use FBFM40, but several exploratory ablation models do; those ablations are retained only as diagnostics in this revision. The current M5 selected core should also be described as a physics-guided selected core, not as an independently tuned final model unless pre-specification or nested model selection is documented.

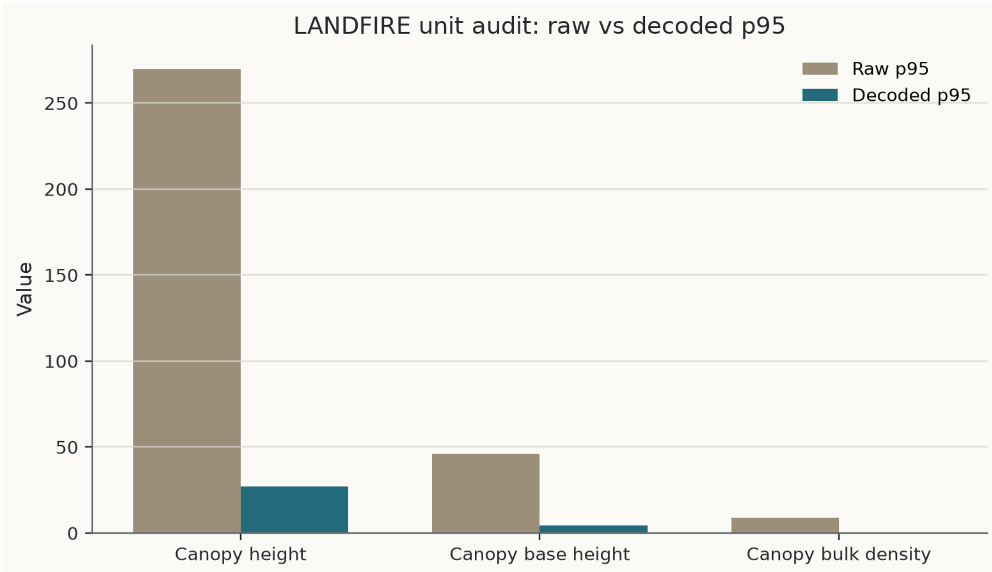

Figure 9: LANDFIRE unit audit for canopy height, canopy base height, and canopy bulk density.

Table 6: Method audit and current interpretation limits.

Item	Current status	Manuscript interpretation
Temporal holdout	2017–2021 train, 2022–2023 test	Useful retrospective check; not prospective forecasting proof.
L2 logistic comparison	Nearly identical discrimination to Bayesian Laplace	Bayesian value is posterior decision quantities.
Brier skill	Slightly negative versus climatology	Do not claim calibration superiority.
Segment dependence	Preliminary segment-cluster bootstrap only	Spatial block validation still required.
Stan MCMC	Initial pilot consistent with Laplace point metrics	Full diagnostics and posterior predictive checks required.
LANDFIRE time matching	LF2024 used for 2017–2023 panel	Time-matched LANDFIRE sensitivity required.

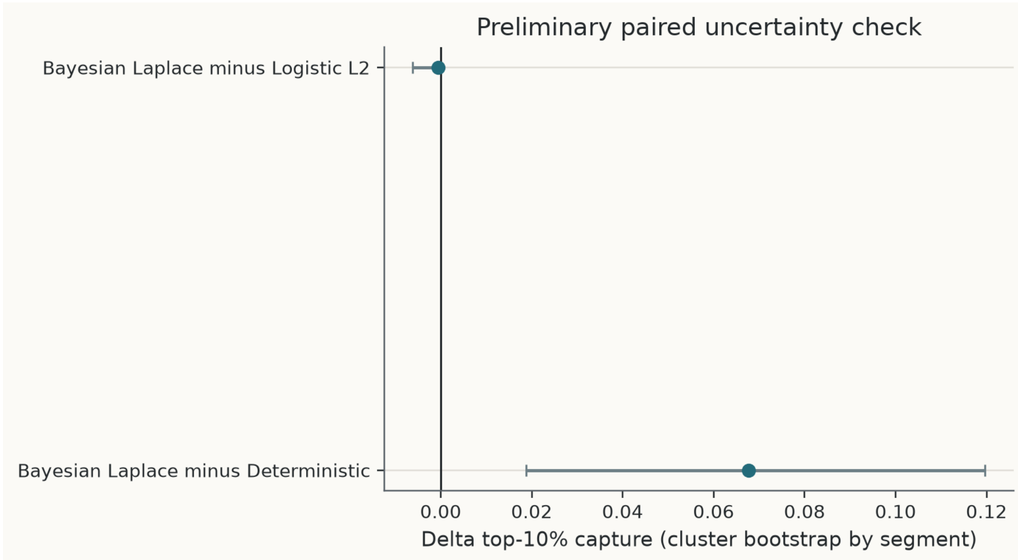

Figure 10: Preliminary paired uncertainty check from segment-cluster bootstrap. This does not replace spatial block validation.

6 Discussion

The main result is bounded. Public data can support a reproducible, screening-level framework for retrospective external wildfire exposure ranking of transmission-line receptor assets. The output is a ranked exposure screen, not a statement about ignition causality, equipment condition, outage, damage, or responsibility.

The temporal-holdout results do not show broad Bayesian classification superiority. L2 logistic and Bayesian Laplace have nearly identical ROC-AUC, PR-AUC, Brier score, and top-decile capture. This is consistent with the relationship between Gaussian-prior logistic regression and L2 penalization. The Bayesian contribution is instead decision-facing: posterior draws convert a fitted exposure model into posterior exceedance probability and posterior top-decile rank probability. These quantities are better aligned with rare-event triage than a single deterministic score, because they express uncertainty in threshold and rank decisions.

The deterministic comparator remains useful because it is transparent and easy to audit. Its limitation is that it gives one ranking without posterior rank uncertainty. Flexible tree-based ML baselines do not improve this temporal holdout, but this result should be interpreted cautiously. It supports careful feature design and temporal validation, not a general rejection of ML.

Existing wildfire risk tools target different questions. Powerline ignition and equipment-failure models generally require private utility data and address grid-to-fire causality. Fire-spread simulators can represent richer physical dynamics but require scenario assumptions and are not automatically calibrated for public-data segment ranking. The present framework occupies a narrower space: public-data fire-to-grid external exposure ranking with a posterior decision layer.

7 Limitations

Several limitations are important for a preprint. First, the pipeline estimates external exposure only. It does not estimate ignition causality, equipment failure, outage probability, damage, or legal responsibility. Second, the CAL FIRE cause field is a perimeter-level attribute; electrical-power overlap is therefore only a diagnostic label. Third, LF2024 static fuel layers may be temporally mismatched with 2017–2023 exposure years; time-matched LANDFIRE sensitivity is required. Fourth, FBFM40 must be encoded categorically or removed from ablations. Fifth, the current label uses a fixed 1 km buffer; 250 m, 500 m, 1000 m, and 2000 m buffer sensitivities should be reported. Sixth, temporal holdout is not spatial block validation. Finally, the Stan run is a pilot; full diagnostics, longer chains, convergence summaries, and posterior predictive checks are still needed.

8 Conclusion

This paper presents a public-data framework for retrospective external wildfire exposure ranking of transmission-line receptor assets. The workflow builds a 2017–2023 segment-year panel, uses cause-screened exposure labels, and compares deterministic, Bayesian, and machine-learning baselines under a 2022–2023 temporal holdout. Bayesian and L2 logistic point-prediction performance is similar. The Bayesian formulation adds posterior exceedance and posterior top-decile rank probabilities, with posterior rank probability emphasized as the main decision quantity under rare-event calibration limits. The framework should not be interpreted as a powerline ignition, equipment-failure, outage, damage, or responsibility model.

A Covariate Diagnostics

The following maps document the public covariate layers used to construct dynamic weather, terrain, and fuel-structure predictors. They are included as quality-control and covariate-diagnostic figures rather than as separate scientific claims.

B Reproducibility Notes

All data processing, feature construction, model fitting, validation, and figure generation are scripted in the project repository. Public-source citations are still marked as TODO placeholders until a bibliography is added.

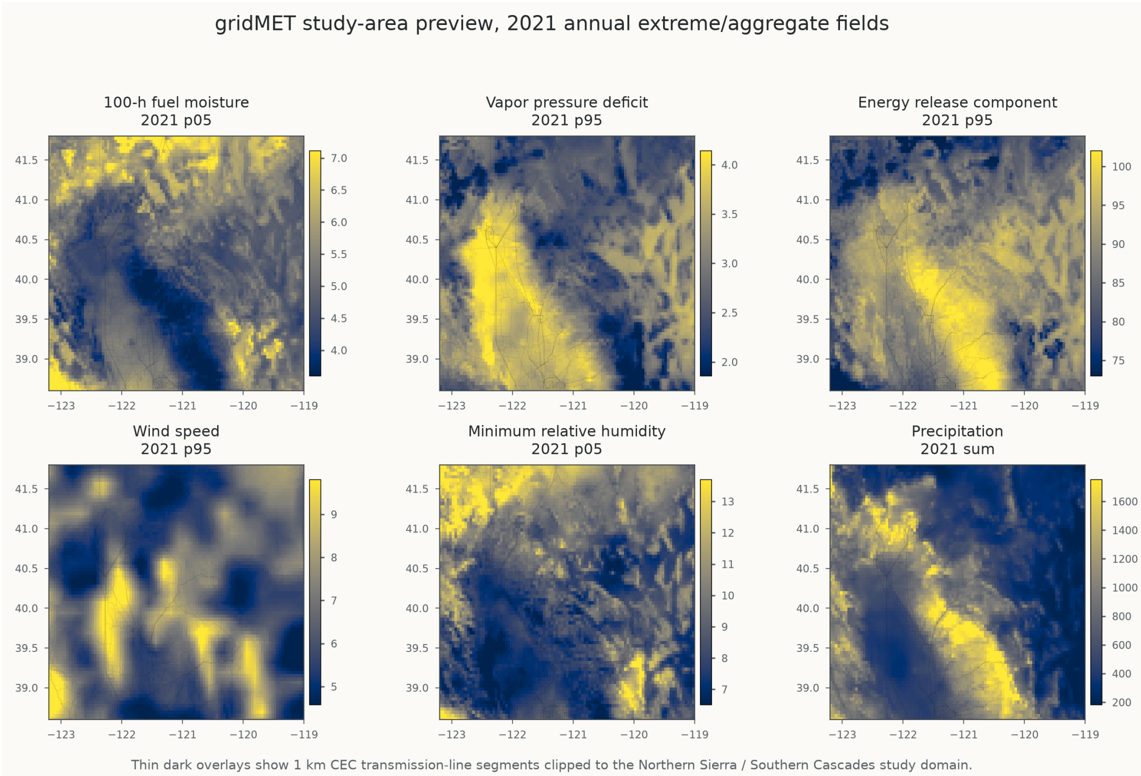

Figure 11: gridMET annual fire-weather covariate layers for 2021.

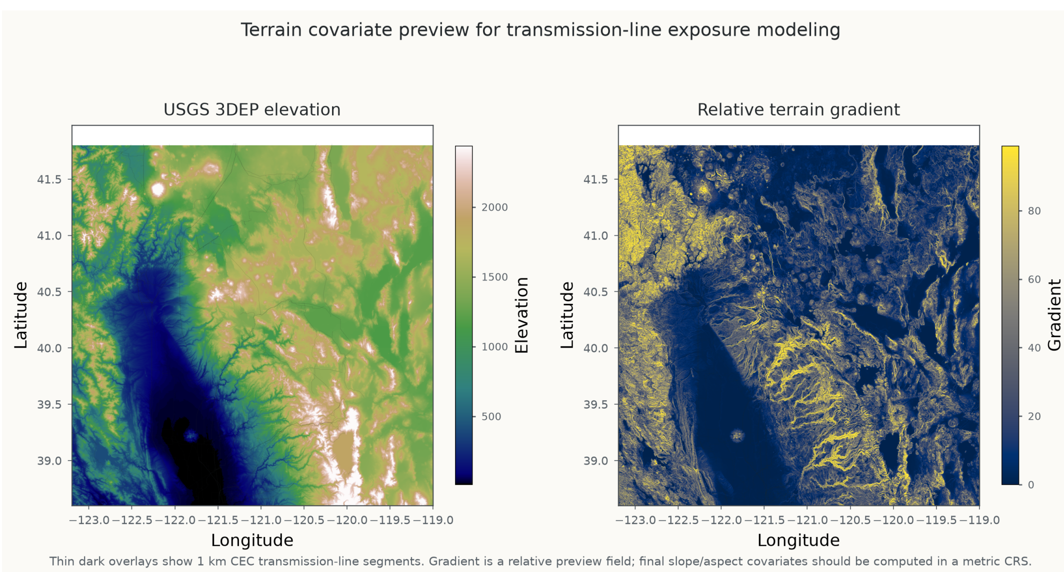

Figure 12: USGS 3DEP terrain covariate layers.

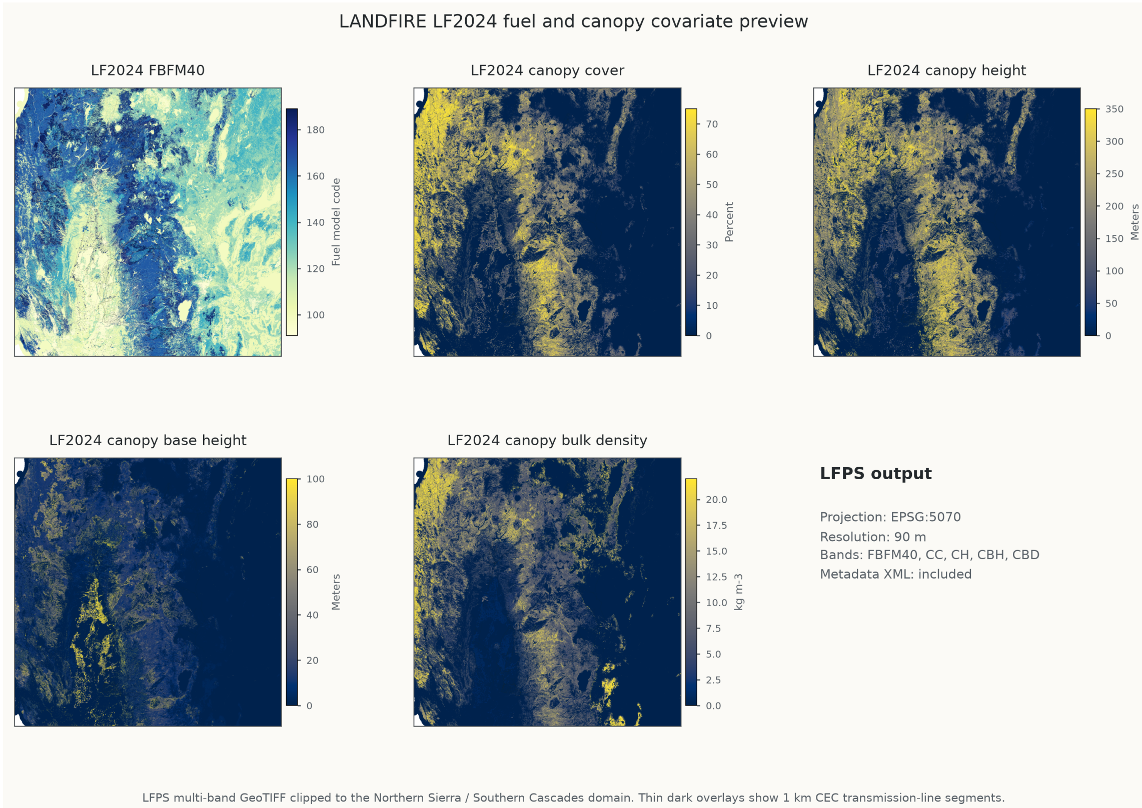

Figure 13: LANDFIRE LF2024 fuel and canopy covariate layers.

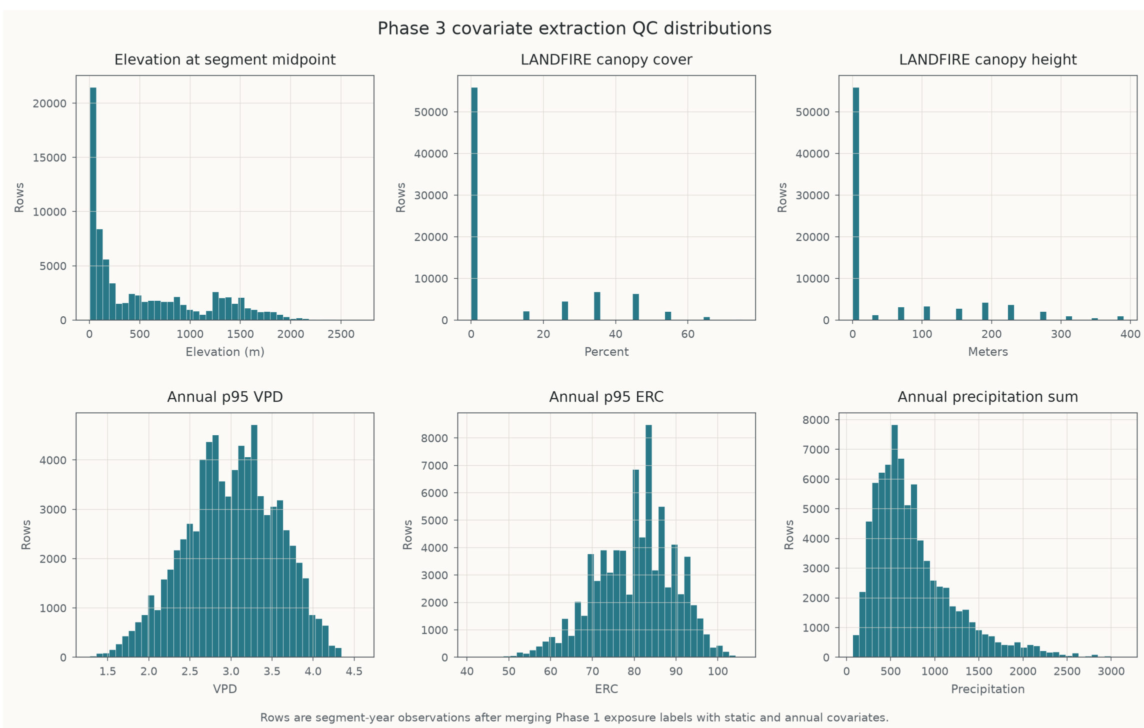

Figure 14: Covariate distribution diagnostics.

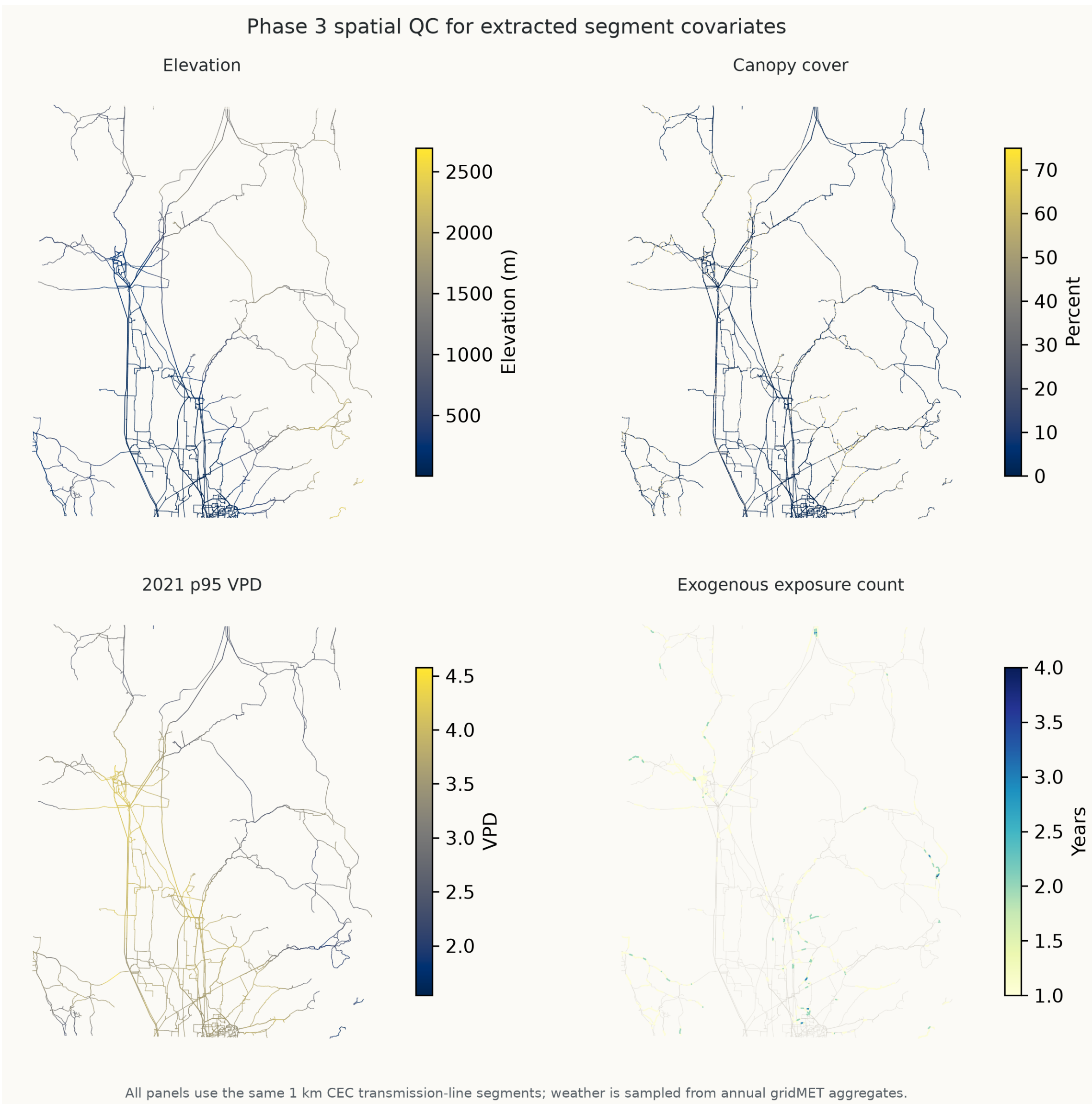

Figure 15: Spatial covariate quality-control maps.